

CPT204 Mock Exam - Question 13

OnlineCourse

Given the following interface:

```
interface Payable {  
    double getPaymentAmount();  
}
```

Q13 OnlineCourse: constructor assigns all fields. getPaymentAmount: if baseFee<0 return 0; amount=baseFee + level surcharge (0/100/200); if hasCertificate add 50; return amount.

Complete the class OnlineCourse that implements Payable.

An online course has:

- courseName
- baseFee
- level
- hasCertificate

The course level surcharge rules are:

- BEGINNER: +0
- INTERMEDIATE: +100
- ADVANCED: +200

If hasCertificate is true, add an extra 50 after adding the level surcharge.

If baseFee is negative, getPaymentAmount() should return 0.

Answer:

```
interface Payable {  
    double getPaymentAmount();  
}  
  
class OnlineCourse implements Payable {  
    private String courseName;  
    private double baseFee;  
    private Level level;  
    private boolean hasCertificate;  
  
    public enum Level {  
        BEGINNER, INTERMEDIATE, ADVANCED  
    }  
  
    public OnlineCourse(String courseName, double baseFee,  
                        Level level, boolean hasCertificate) {  
        // TODO: Please complete this method.  
    }  
  
    @Override  
    public double getPaymentAmount() {  
        // TODO: Please complete this method.  
    }  
}
```

CPT204 Mock Exam - Question 14

KeywordCounter

Given the following interface:

```
interface Countable {  
    int getKeywordCount();  
}
```

Q14 KeywordCounter: constructor sets text, keywords=new HashSet<>(keywords), wordCounts=new HashMap<>(), then countWords(). countWords splits text and increments map counts. getKeywordCount sums counts for keywords; if text null/empty return 0

Complete the class KeywordCounter that implements Countable.

A KeywordCounter has:

```
private String text;  
private HashSet<String> keywords;  
private HashMap<String, Integer> wordCounts;
```

The class counts how many times selected keywords appear in a piece of text.

The input text will contain lowercase words separated by single spaces only. There will be no punctuation.

Rules:

- The constructor should initialise all properties.
- Store all given keywords in the HashSet.
- Count all words in the text using the HashMap.
- getKeywordCount() should return the total number of times any keyword appears in the text.
- If text is null or empty, return 0.

Answer:

```
import java.util.*;  
  
interface Countable {  
    int getKeywordCount();  
}  
  
class KeywordCounter implements Countable {  
    private String text;  
    private HashSet<String> keywords;  
    private HashMap<String, Integer> wordCounts;  
  
    public KeywordCounter(String text, Collection<String> keywords) {  
        // TODO  
    }  
  
    private void countWords() {  
        // TODO  
    }  
  
    @Override  
    public int getKeywordCount() {  
        // TODO  
    }  
}
```